# Yashank Patidar

(+91) 9165513163 | yashankpatidar21@gmail.com | LinkedIn: Yashank | GitHub: yashank21

## Professional Summary

AI/ML Engineer and Software Engineering Intern with hands-on experience across Retrieval-Augmented Generation (RAG), generative modeling (GANs), multimodal transformers, and backend systems optimization. At Juniper Networks, delivered a RAG chatbot with 92% response relevance, reduced computation overhead by 30%, and cut testing-pipeline latency by 14%. Built two independent research-grade projects: a GAN-based 2D-to-3D voxel reconstruction system and a DeBERTa + ViT multimodal meme classifier (68.28% accuracy). Proficient in Python, C++, SQL, PyTorch, TensorFlow, LangChain, and Scikit-learn, with a strong foundation in core Computer Science fundamentals. GATE CS 2024 qualifier (Score: 581). Currently pursuing MTech in Information Technology at NIT Karnataka.

## Education

**National Institute of Technology Karnataka, Surathkal** — **CGPA: 7.69/10**
*MTech, Information Technology* — *2024 – 2026*

**Medi-Caps University Indore, MP** — **CGPA: 8.89/10**
*BTech, Computer Science Engineering* — *2020 – 2024*

## Experience

**Juniper Networks** — **June 2025 – May 2026**
*Software Engineering Intern*

- Engineered a Retrieval-Augmented Generation (RAG) chatbot for domain-specific query resolution, achieving **92% response relevance accuracy** through optimized retrieval and context-ranking logic.
- Reduced system computation overhead by **30%** by redesigning resource allocation algorithms, improving backend efficiency and scalability under load.
- Re-engineered the Unit Test (UT) generation pipeline, cutting execution latency from **350ms to 300ms (a 14% reduction)**, improving CI throughput, system responsiveness, and overall testing efficiency.

## Projects

**2D Image to 3D Voxel Reconstruction using GANs** | *Python, TensorFlow, Computer Vision, GANs*

- Designed and trained a Generative Adversarial Network (GAN) to reconstruct 3D voxel models from single 2D images, using the ModelNet10 dataset spanning **10 object categories**.
- Built an end-to-end data pipeline covering 2D rendering from .off mesh files, voxelization, preprocessing, and paired-dataset generation, enabling supervised GAN training.
- Architected CNN and 3D-CNN-based Generator and Discriminator networks incorporating normalization and batch training, enabling automated 3D shape synthesis from single-view images.

**Multimodal Meme Classification using Transformers** | *Python, PyTorch, DeBERTa, ViT, NLP, OCR*

- Developed a multimodal classification system combining text and image embeddings (DeBERTa + Vision Transformer) to categorize memes into Hero, Villain, Victim, and Other classes.
- Implemented OCR-based text preprocessing, Named Entity Recognition (spaCy), and feature fusion to generate unified multimodal representations for classification.
- Trained a PyTorch-based neural network on COVID-19 and political meme datasets, achieving **68.28% accuracy** on the held-out test set.

## Technical Skills

**Programming Languages:** Python, C++, SQL
**Generative AI / LLM:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain
**AI/ML & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, GANs, Transformers (DeBERTa, ViT)
**Computer Vision & NLP:** Computer Vision, Natural Language Processing, OCR, Named Entity Recognition
**Data & Backend:** Pandas, NumPy, RAG-based backend integration
**Developer Tools:** Git, VS Code, Jupyter Notebook, Google Colab
**Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS

## Achievements & Certifications

- **GATE CS 2024** – Score: 581 (national-level CS/IT graduate aptitude exam).
- **CCNA Certified** – Networking Fundamentals (Cisco).